CIVIL ENGINEERING

IV B.Tech – I Semester

Course Code	**FINITE ELEMENT METHODS**	L	T	P	C
23A01701		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the fundamental principles of the Finite Element Method (FEM) and its applicationsin structural analysis.
2. **Apply** the concepts of elasticity, stress-strain relationships, and displacement functionsin FEM.
3. **Develop** finite element formulations for1D, 2D, and 3D elements.
4. **Analyze** bar, beam, and plane stress/strain problems using shape functions and stiffness matrices.
5. **Implement** solution techniques such as numerical integration, static condensation, and element assembly.

Course Outcomes (COs):

Upon successful completion of this course, students will be able to:

1. **Explain** the basic principles, advantages, and limitations ofFEMin engineering applications.
2. **Derive** shape functions and stiffness matrices for 1D bar and beam elements.
3. **Analyze** continuous beams and plane stress/strain problems using FEM.
4. **Develop**isoparametric formulation for 2D elements such as CST and LST.
5. **Implement** solution techniques for assembling elements and solving FEM equations.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	2	2	-	-	-	-	-	-	2	3	3
CO -2	3	3	3	3	3	-	-	-	-	-	-	2	3	3
CO -3	3	3	3	3	3	-	-	-	-	-	-	2	3	3
CO -4	3	3	3	3	3	-	-	-	-	-	-	2	3	3
CO -5	2	2	2	2	3	-	-	-	-	1	-	2	2	2

UNIT – I

Introduction to Finite Element Method – Basic Equations in Elasticity Stress – Strain Equation – Concept of Plane Stress – Plane Strain Advantages and Disadvantages of FEM. Element Shapes – Nodes – Nodal Degree of Freedom Displacement Function – Natural Coordinates – Strain Displacement Relations.

UNIT – II

Lagrangian– Serendipity Elements – Hermite Polynomials – Regular, Irregular 2 D &3D – Element – Shape Functions Up to Quadratic formulation. Finite Element Analysis (FEA) of – One Dimensional Problems – Bar Element – Shape Functions Stiffness Matrix – Stress – Strain Relation

UNIT – III

FEA Beam Elements – Stiffness Matrix - Shape Function– Analysis of Continuous Beams.

UNIT – IV

FEA Two-Dimensional Problem – CST – LST Element – Shape Function – Stress – Strain.

Isoparametric formulation – Concepts of, Isoparametric Elements for2D Analysis - Formulation of CST Element.

UNIT – V

Solution Techniques: Numerical Integration, Static Condensation, Assembly of Elements and Solution Techniques for Static Loads.

TEXT BOOKS:

1. A first coursein Finite Element Method by Daryl L. Logan, 5th Edition, Cengage Learning India Pvt. Ltd.
2. Introduction to finite Elements in Engineering by Tirupathi R. Chandrupatla, and Ashok D. Belegundu, Prentice Hall of India

REFRENCE BOOKS:

1. Finite Element Aanalysis by P. Seshu, PHI Learning Private Limited
2. Concepts and applications of Finite Element Analysis by Robert D. Cook et al., Wiley India Pvt. Ltd.
3. Applied Finite Element Analysis by G. Ramamurty, I.K. International Publishing House Pvt. Ltd.

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/105/105105041/#
https://archive.nptel.ac.in/noc/courses/noc22/SEM1/noc22-me43/